

Bayesian Statistics (47-747)
Homework #1
60 points total

Please provide your R code for all relevant problems.

Problem 1 (10 points):

For this problem you will use the Beta distribution. Suppose that we toss a coin N times and observe that the coin comes up with heads S times. We are interested in estimating the probability θ that the coin comes up heads, in other words your data is generated by a binomial process. Assume that your prior follows an arcsine probability density

function:
$$f(\theta) = \frac{1}{\pi\sqrt{\theta(1-\theta)}}$$

(a) Derive the posterior distribution of θ . What is the mean and variance of your posterior distribution?

(b) Using R plot the prior, likelihood, and posterior in a single graph for the following two cases. Provide your R code and your two plots.

- i. $S=1$ and $N=1$
- ii. $S=4$ and $N=6$.

(c) Suppose you observe another sample of information, this time you observe S_1 heads after N_1 trials. Derive the posterior using the previous posterior from part (a) as your new prior.

Problem 2 (10 points):

Suppose you have constructed an econometric model that gives a forecast of annual US GDP for the next year. Since GDP is unknown, we denote it as the random variable θ .

(a) As your prior, you believe your forecast is well described by a normal distribution with a mean of -2% growth and variance of 2.25%. Write out the probability density function with a mean of μ_0 and precision parameter of τ_0 , and plot this prior distribution.

(b) There is another expert who has a more optimistic forecast of US GDP for next year, with a mean of 4% growth and variance of 1.5%. This expert's information is independent of yours. Write out the probability density function with a mean of μ and precision parameter of τ and plot your distribution.

(c) How could we combine the two forecasts from parts (a) and (b)? You do not need to actually compute the distributions, but describe in words how you could combine the information from you and the other expert.

Problem 3 (10 points):

Consider the usual regression model where:

$$y_i = \mathbf{x}_i' \boldsymbol{\beta} + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma^2)$$

Where $\mathbf{y}$ and $\mathbf{X}$ are defined below, and $\boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}$ is 2 x 1 vector.

$$\mathbf{y} = \begin{bmatrix} 1 \\ 4 \\ 3 \\ 8 \\ 10 \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}$$

For each of the following cases, compute the posterior for $\boldsymbol{\beta}$ and σ^2 , including the posterior means and 95% credible intervals. Also create a density plot that overlays the prior, likelihood, and posterior for β_1 for each of the cases.

- (a) Diffuse prior.
- (b) Informative prior in which you think $\boldsymbol{\mu}_0 = [1, 2]$, $\boldsymbol{\Lambda}_0$ is an identity matrix, and $\nu_0 = 5$, $\sigma_0^2 = 0.6$.
- (c) A g-prior where $g_0 = 0.1$ and everything else is the same as part (b)
- (d) A g-prior where $g_0 = 1$ and everything else is the same as part (b)
- (e) A g-prior where $g_0 = 10$ and everything else is the same as part (b)
- (f) Comment on the differences for each of the estimates from parts (a)-(e). How does the prior influence your estimates?

Problem 4 (10 points):

In our first week of lecture, we reviewed several examples of applying the Bayesian approach to different data distribution settings. For each distribution, provide a brief summary (in words) of how you would describe the (i) prior, (ii) data likelihood, and (iii) posterior. Be sure to include your interpretation of the parameters within the distributions:

- (a) Beta-Binomial
- (b) Poisson Data
- (c) Normal Distribution

Problem 5 (20 points):

Please read/skim the following 5 papers (all pdfs are available on Canvas).

1. Rossi & Allenby (1993): “A Bayesian Approach to Estimating Household Parameters”
2. Rossi, McCulloch, & Allenby (1996): “The Value of Purchase History Data in Target Marketing”
3. Montgomery (1997): “Creating Micro-Marketing Pricing Strategies Using Supermarket Scanner Data”
4. Montgomery & Rossi (1999): “Estimating Price Elasticities with Theory-Based Priors”
5. Rossi & Allenby (2003): “Bayesian Statistics and Marketing”

Note: There is a big difference between Empirical Bayes and Hierarchical Bayes (we really only consider the latter to be “real” Bayes)

Use the content of the readings to answer the following discussion questions. Note that the main theme across these papers is modeling household purchase behavior within a particular product category (e.g., different brands of canned tuna or orange juice) using scanner panel data. More credit will be given if you provide detailed citations/references in your response.

- (a) Describe the overall differences between a random effects vs. fixed effects vs. Bayesian approach in estimating, for example, price elasticities using household scanner panel data.
- (b) What types of information from scanner panel data can be used by researchers to model heterogeneity across households? How can this information be incorporated into the models?
- (c) What are some reasons for why the Bayesian approach is particularly useful in modeling heterogeneity across households when using scanner panel data? (Another way of thinking about this question is what limitations of the data does the Bayesian approach address?)
- (d) How do marketing researchers make use of or interpret the “prior” when estimating models of household purchase behavior?
- (e) Describe your understanding of the following two terms in words: hierarchical Bayes and MCMC. Why do we need to use MCMC when estimating hierarchical Bayesian models, in the context of household purchase behavior?